

THE UNITED REPUBLIC OF TANZANIA

PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT



MPWAPWA DISTRICT COUNCIL

FORM THREE MID-TERM EXAMINATION

031

PHYSICS 1

TIME: 3:00 Hours

SEPTEMBER 2026

INSTRUCTIONS

1. This paper consists of sections A, B, C with total of **eleven (11)** questions.
2. Answer **all** questions in section A and B and **two (2)** questions in section C
3. Section A carries **sixteen (16) marks**, section B **fifty-four (54) marks** and section C **thirty (30) marks**.
4. Communication devices and any unauthorized materials are not allowed in the examination room
5. Non programmable calculators and mathematical tables may be used
6. Write your **examination number** on every page of your answer booklet
7. Where necessary the following constants may be used:
 - (i) Acceleration due to gravity, $g = 10 \text{ m/s}^2$
 - (ii) Density of water = 1 g/cm^3 or 1000 kg/m^3
 - (iii) Pie $\pi = 3.14$
 - (iv) Specific heat capacity of water = $4200 \text{ J/kg}^\circ\text{C}$

SECTION A: (16 MARKS)

Answer **all** questions in this section

1. For each of items (i) – (x) choose the correct answer from among the given alternatives and write its letter besides the item number in the answer booklet provided.

(i) When locating the image formed by a concave mirror, any two rules are enough to locate the image. Which of the following rules can be used?

1. An incident ray that travels parallel to the principal axis of the mirror is reflected through the centre of curvature
2. An incident ray that passes through the centre of curvature is reflected parallel to the principal focus.
3. An incident ray that passes through the focal point is reflected parallel to the principal axis.
4. An incident ray not parallel to the principal axis, but strikes the pole of the mirror, is reflected back making the same angle with the principal axis.

A. 1 only B. 1 and 3 C. 2 and 3 D. 3 and 4 E. 4 only

(ii) Which of the following statements is **NOT** true about a simple microscope and a compound microscope?

- A. Angular magnification of a simple microscope is also known as the magnifying power.
- B. The image is formed at infinity when the position of the magnifying glass or the object itself is adjusted so that the object is at focal point of the magnifying glass.
- C. Both use plano convex lens system that enlarge the image.
- D. Compound microscope forms two images during its operation.
- E. Compound microscope provides larger magnification than simple microscope.

(iii) When Juma was measuring the density of a liquid A by using a hydrometer, he found that the hydrometer sank more in liquid A than in water when the same container was used for both. What was the best conclusion from his experiment?

- A. Upthrust provided by water is greater than the upthrust provided by liquid A.
- B. The weight of liquid A displaced is greater than the weight of water displaced by hydrometer.

- C. The density of liquid A is greater than the density of water
 - D. The density of water is greater than the density of liquid A
 - E. The hydrometer was partially immersed in water while completely immersed in liquid A.
- (iv) Small insects such as water striders and pond skaters are able to walk on water because:
- A. The cohesive force between liquid molecules is strong
 - B. The adhesive force between liquid molecules is weak.
 - C. The adhesive force between liquid molecules is strong.
 - D. The weight of small insects is equal to the weight of water displaced by the small insects.
 - E. The weight of small insects is less than the upthrust provided by water.
- (v) Which of following is **NOT** a reason why physicists prefer the aneroid barometer over the Fortin barometer?
- A. Fortin barometer uses mercury which very expensive and toxic to use.
 - B. Fortin barometer must be mounted in a vertical position
 - C. Aneroid barometer has a metal chamber contains non-toxic liquid which helps in registering the pressure.
 - D. Fortin barometer has a length of about 1 m.
 - E. Aneroid barometer is highly portable.
- (vi) A group of five form three students at Godegode secondary school was performing an experiment on relative humidity by using a wet and dry bulb hygrometer. They found that the temperatures measured by both dry and wet bulb thermometers were 25 °C and 10 °C respectively and each student gave his opinion. Who do you think was correct?
- A. The air is saturated with water vapour.
 - B. The humidity is very high in air.
 - C. The vapour partial pressure is equal to the atmospheric pressure.
 - D. The rate of evaporation is higher than the rate of condensation.
 - E. The formation of dew is favourable in this condition.
- (vii) What value of capacitor could be used to replace a set of 5 μF , 10 μF and 15 μF capacitors connected in series?

A. $30\ \mu\text{F}$ B. 0.37 C. $0.37\ \mu\text{F}$ D. $2.73\ \mu\text{F}$ E. $25\ \mu\text{F}$

(viii) When a teacher was teaching about the pulley as a simple machine, he asked students to explain the difference between a single movable pulley and a single fixed pulley. Which student gave the correct difference?

- A. A single fixed pulley multiplies the effort applied while a single movable pulley changes the direction of the applied force.
- B. A single fixed pulley has a VR of greater than 1 while a single movable pulley has a VR of less than 1.
- C. A single fixed pulley is affected by friction while a single movable pulley is not affected by friction.
- D. A single fixed pulley changes the direction of the applied force while a single movable pulley multiplies the effort applied.
- E. A single fixed pulley is used to carry loads horizontally while a single fixed pulley carries loads vertically.

(ix) Suppose you exert an upward force of 10 N on a 3 kg object. What will be the object's acceleration?

- A. $6.6\ \text{m/s}^2$ (downward).
- B. $3.3\ \text{m/s}^2$ (downward).
- C. $3.3\ \text{m/s}^2$ (upward)
- D. $6.6\ \text{m/s}^2$ (upward)
- E. $10\ \text{m/s}^2$ (downward)

(x) A gas occupies a volume of $20\ \text{cm}^3$ at 27°C and at normal atmospheric pressure. Calculate the final temperature if the gas is heated resulting in a change of volume of $1.88\ \text{cm}^3$.

- A. 28.2 K B. 29.5 K C. 328 K D. 2.54 K E. 40 K

2. Match the light colours in list A with their corresponding properties in list B by writing the letter of the correct response beside the corresponding item number in the answer booklet provided.

LIST A	LIST B
(i) Blue colour	A. Is a complementary colour of cyan
(ii) Red colour	B. Is a complementary colour of magenta
(iii) Green colour	C. Is a complementary colour of yellow
(iv) Yellow colour	D. It absorbs blue
(v) Magenta colour	E. It absorbs green
(vi) Cyan colour	F. It absorbs red

SECTION B: (54 MARKS)

Answer **all** questions in this section

3. (a) A ray of light strikes a plane mirror such that the angle with the mirror is 20° . What is the value of the angle between the incident ray and the reflected ray? **(4 marks)**
- (b) A small quantity of powdered charcoal is placed in a 100 cm^3 density bottle and the total mass is measured to be 255 g. If the mass of empty density bottle is 20 g and when the bottle is filled with water, the total mass is 289 g, what is the density of the powdered charcoal? **(5 marks)**
4. (a) A mass 3 kg hangs at the end of a string. Find the horizontal force needed to pull the mass sideways until the string is at 30° to the vertical. Find also the tension in the string at the final position. **(6 marks)**
- (b) Give the difference between electromotive force and potential difference as used in current electricity. **(3 marks)**
5. (a) With aid of a diagram briefly explain how land breeze occurs. **(5 marks)**
- (b) A copper pipe of length 100 cm at 15°C increases its length by 0.15% when steam at 100°C passes through. Find the coefficient of linear expansivity of copper.

(4 marks)

6. (a) A uniform metre rule is pivoted at its centre O with a knife edge. A 20 g mass is placed at the 10 cm mark and a 50 g mass at the 40 cm mark. At what mark must a second 50 g mass be placed for the system to be in rotational balance? (5 marks)
- (b) Can an object at rest be in a state of equilibrium? Explain your answer. (4 marks)
7. (a) The pulley system shown in Figure 1 is to be used to raise a load of 150 N.

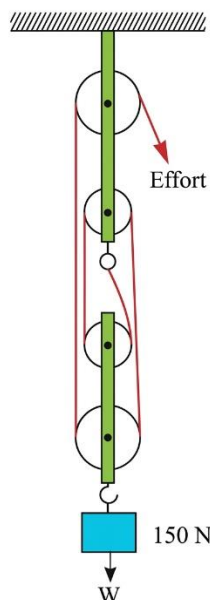


Fig. 1

- (i) Ignoring friction and the weight of the pulleys, what force is required to raise the load at a constant speed? (3 marks)
- (ii) If the cable (where the effort is applied) is pulled down 60 cm, how far will the object rise? (2 marks)
- (b) Briefly explain how the fixed points of a thermometer are obtained. (4 marks)
8. (a) Table below shows the dew point in four Dodoma's districts which have the same air temperature of 20°C at 7:00 a.m.

District	Dew point (°C)
Mpwapwa	1
Kongwa	12
Bahi	5
Kondoa	4

- (i) In which district is the relative humidity the highest? Explain. (2 marks)

- (ii) In which town is the relative humidity the lowest? Explain. **(2 marks)**
- (b) Most of the people use the term boiling instead of evaporation because they think these terms mean the same thing. As a form three student give the differences between boiling and evaporation. Give five points. **(5 marks)**

SECTION C (30 Marks)

Answer **two (2)** questions from this section

9. (a) How much thermal energy is needed to turn 3.0 kg of water at 50 °C into steam at 100 °C? The specific heat capacity of water is 4200 J/kg °C. Specific latent heat of vapourisation of water is 2260 kJ/kg. **(5 marks)**
- (b) A brass cylinder of mass, m_b was heated to 100 °C and then quickly transferred into a thin aluminium can of negligible heat capacity, containing 150 g of paraffin at 11 °C. If the final steady temperature of the paraffin was 20 °C, determine the value of m . (Specific heat capacity of brass is 320 J/kg°C and specific heat capacity of paraffin is 2200 J/kg°C). **(5 marks)**
- (c) With aid of a diagram explain briefly how calorimeter can reduce heat loss. **(5 marks)**
10. (a) An air bubble at the bottom of a lake 90 m deep has a volume of 1.5 cm³. What will be its volume just below the surface if the atmospheric pressure is equivalent to 10 m of water? **(5 marks)**
- (b) A 24 kg box is released on a 30° inclined plane and accelerates at 0.25 m/s² down the plane. Find the frictional force impeding its motion. **(5 marks)**
- (c) A rectangular box of mass 10 kg rests on an incline. Let the coefficient of static friction be 0.55 and that of kinetic friction be 0.25. At what angle will the box begin to slide? **(5 marks)**
11. (a) Explain why a glass container with walls is more likely to crack than one with thin walls when a very hot liquid is poured into each of the glasses. **(5 marks)**
- (b) An object of 4 cm height is placed at 6 cm in front of a concave mirror of radius of curvature 24 cm. Find the position, height, magnification and nature of the image. **(5 marks)**
- (c) A swimming pool appears to be 1.5 m deep. If the refractive index of water is 1.3, determine the real depth of the pool. **(5 marks)**